Enterprise Reporting & Automation Strategy Using Microsoft Tools

Case Study Response: ShopEase - Azure Compute and Networking Components

Objective: Design an Azure-based architecture that keeps the e-commerce site highly available, scalable, secure, and cost-efficient during traffic surges while protecting the internal database.

1. Problem Analysis

ShopEase is a fast-growing e-commerce company whose on-premises setup struggles during peak sale events. The main issues are poor performance under heavy load, limited scaling, downtime risk during hardware failures, and insecure exposure of application components that should remain private.

2. Business and Technical Requirements

- Keep the web and database tiers available even during maintenance or hardware failure.
- Distribute customer traffic evenly so no single server becomes overloaded.
- Scale compute resources automatically during demand spikes and reduce them when traffic falls.
- Protect internal communication between application and database tiers.
- Provide disaster recovery and failover support across regions.

3. Proposed Azure Solution

The recommended solution combines Azure compute and networking services to build a resilient two-tier architecture. The web application runs on Azure Virtual Machines, availability is improved through Availability Sets or VM Scale Sets, traffic is distributed using Azure Load Balancer, secure communication is maintained through a Virtual Network and Network Security Groups, and regional failover is handled with Azure Traffic Manager.

4. Azure Services and Their Roles

Azure Virtual Machines: Host the ShopEase web application with full control over operating system, software stack, and configuration.

Availability Set: Places VMs across fault domains and update domains so maintenance or hardware failure does not bring the whole application down.

Virtual Machine Scale Set: Automatically increases or decreases the number of VM instances based on CPU usage or incoming requests.

Azure Load Balancer: Distributes traffic across healthy VM instances and supports both public and internal load balancing.

Azure Virtual Network: Creates a private network for communication between the web tier and database tier.

Network Security Groups: Restrict inbound and outbound traffic, for example allowing only HTTPS from the internet and tightly controlling database access.

Azure Traffic Manager: Provides geo-redundancy and failover by routing users to the primary region and switching to a secondary region when needed.

Azure Backup: Protects VM data and configurations for recovery after accidental loss or system failure.

5. Suggested Architecture Flow

1. Users access the ShopEase website over HTTPS.
2. Azure Traffic Manager directs users to the active region.
3. A public Azure Load Balancer distributes incoming requests across web VMs.
4. The web tier runs inside a VM Scale Set for automatic scaling.
5. Web VMs communicate privately with the database tier through a VNet.
6. NSGs allow only required ports and block unnecessary public exposure.
7. Azure Backup and a secondary region support recovery and failover.

Users → Traffic Manager → Public Load Balancer → VM Scale Set / Web VMs → Internal Load Balancer → Database Tier

6. Step-by-Step Implementation

8. Create a Virtual Network with separate subnets for the web tier and database tier.
9. Deploy the initial Azure Virtual Machines for the web application.
10. Place the VMs in an Availability Set to improve uptime during maintenance and failures.
11. Create a VM Scale Set and configure autoscaling rules such as scale out when CPU usage rises above a threshold.
12. Deploy a public Azure Load Balancer and configure health probes for the web tier.
13. Deploy an internal load balancer for private communication between application and database components.
14. Apply Network Security Groups to allow HTTPS to the web tier and restrict database access to internal traffic only.
15. Enable Azure Backup for VM protection and recovery.
16. Configure Azure Traffic Manager with primary and secondary regional endpoints for disaster recovery.
17. Test high traffic and failover scenarios to confirm autoscaling, health checks, and recovery behavior.

7. Benefits of the Proposed Design

- High availability because workloads are spread across multiple fault and update domains.
- Elastic scalability because VM Scale Sets add or remove instances automatically.
- Better customer experience because traffic is balanced across healthy servers.
- Lower cost because resources scale according to actual demand instead of staying oversized.

- Stronger security because the database tier remains private inside the VNet.
- Business continuity because Traffic Manager and backups support failover and recovery.

8. Final Recommendation

ShopEase should adopt a two-tier Azure architecture built on Virtual Machines, Availability Sets, VM Scale Sets, Load Balancers, VNets, NSGs, Azure Backup, and Traffic Manager. This design directly addresses the company's current issues of downtime, poor scalability, and security exposure. It also creates a future-ready platform that can handle peak sales periods with better reliability, lower operational effort, and improved business continuity.